Proceedings of Machine Learning Research vol vvv:1–13, 2027

Eddies: Dynamical Action Tokens with Certified Composition

Aditya Narendra ADITYA.NARENDRA@MBZUAI.AC.AE
Grik Tadevosyan GRIK.TADEVOSYAN@MBZUAI.AC.AE
Abdalla Swikir ABDALLA.SWIKIR@MBZUAI.AC.AE
Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE

Abstract

Action tokens describe intended behavior but do not quantify tracking accuracy or error propagation across skills. We introduce Eddies, action tokens that combine a primitive bound to a task frame, an identified robot-specific closed-loop model, and a conformal residual envelope. Our composer, Streamer, propagates tracking-error bounds across hand-overs, including the change in equilibrium; for fixed, stable models and exchangeable episodes this yields probabilistic tracking certificates for entire Eddie streams. Each Eddie gates its own commands to respect a tracking tolerance whenever its residual envelope holds. Online adaptation removes the need for task-specific calibration and controls the long-run miss rate. On ten simulated tasks across four robots, in a second physics engine and on a real Franka, Eddies produce tighter bounds under a common certification procedure than global models, per-primitive models and learned tokenizers. Certificates are violated in 5 of 2,000 simulated test episodes and in none of 50 real-robot cube-stacking runs, and transfer experiments show that certificates depend on the executing system's dynamics.

Keywords: Action tokens, certified composition, conformal prediction, system identification, manipulation

1. Introduction

Vision-language-action models act through tokens: binned or compressed codes that decode to reference commands (Brohan et al., 2023; Pertsch et al., 2025; Shafiullah et al., 2022; Lee et al., 2024), or primitives that a language model strings together (Ahn et al., 2022). This treats an action like a word, but an action is not a word: its effect depends on where the robot starts and which robot runs it, and the next action starts wherever the previous one left the robot. No action token bounds how well the robot will track it or how the error propagates to the next token.

We make the token carry this information (Fig. 1). An *Eddie* (Embedding-space Discrete Dynamical Instruction with Envelope) is a primitive bound to a task frame, such as grasping a peg, plus a model of how this robot tracks the primitive and an envelope on the model's error. Our composer, Streamer, chains Eddies into an *Eddie stream*: every Eddie gates its own next command against the tracking tolerance, stopping the stream when the check fails, and the error bounds chain through hand-overs into a bound over the whole stream. The certificate covers tracking only; whether a skill succeeds is left to symbolic contracts and to the supervisor that picks the next Eddie.

Our contributions are as follows. (i) Eddies: action tokens that carry an identified closed-loop model and a conformal envelope, built for each robot from its own primitives (Section 4). On the Franka, splitting the dynamics per token roughly halves the certified tube of one global model or one model per primitive, and a skill's certificate can change with the robot and the physics engine that run it. (ii) A chaining rule that carries the actual bound across hand-overs, including the jump

© 2027 A. Narendra, G. Tadevosyan & A. Swikir.

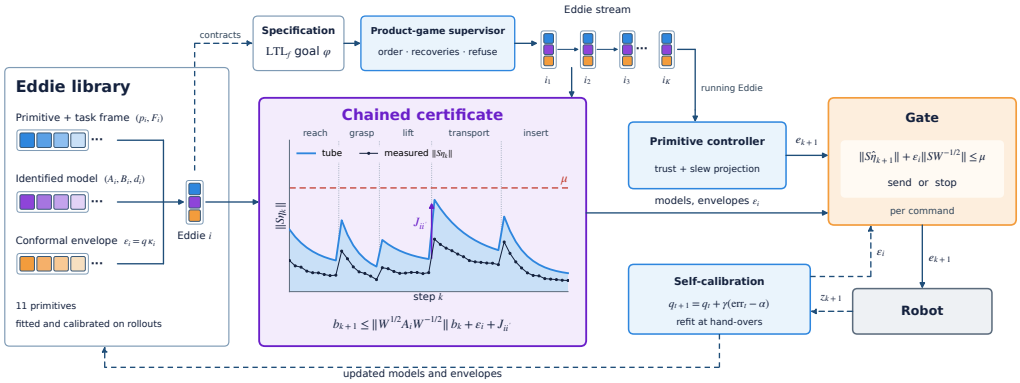


Figure 1: Streamer chains certified Eddies into an Eddie stream; each Eddie gates its own commands and is updated from the closed loop.

between equilibria, which certifies hand-overs even where funnel containment fails. (iii) A self-calibrating gate carried by each Eddie, which checks the Eddie's own next command and tunes itself from closed-loop feedback, with a one-step look-ahead guarantee when the envelope holds and a long-run miss-rate bound (Theorem 1). (iv) A comparison with learned action tokenizers under the same certificate, in which Eddies certify more tightly on all four robots. We evaluate ten tasks on four robots in MuJoCo, three of them also in Isaac Sim (Section 5), and cube stacking on a real Franka (Section 5.5); proofs, identification details and per-task tables are in the supplement. Code, data and videos: https://github.com/Ozzey/EDDIES.

2. Related Work

Action tokenizers have moved from per-dimension bins (Brohan et al., 2023) to richer codes based on k-means, frequency transforms, vector quantization and B-splines (Shafiullah et al., 2022; Pertsch et al., 2025; Lee et al., 2024; Mete et al., 2024; Zhou et al., 2025). These approaches represent and generate actions; none models how the robot tracks them. Primitive libraries (Dalal et al., 2021; Nasiriany et al., 2022) and language-model planners over primitives (Ahn et al., 2022) name skills without a tracking model.

Conformal prediction gives coverage guarantees for prediction and reachability analysis (Lindemann et al., 2023; Cleaveland et al., 2024; Lindemann et al., 2025); combined with contraction analysis and Koopman models it yields probabilistic tracking-error bounds (Hirano and Tsukamoto, 2026; Wei et al., 2025), and Nath et al. (2026) reuse one Koopman model and one calibration across reference trajectories. Eddies attach models and envelopes to individual tokens and compose the bounds across hand-overs, and adaptive conformal inference (Gibbs and Candès, 2021) lets the envelopes adapt from execution feedback when the task changes. Runtime monitors detect failures of generative policies from action consistency and task progress (Agia et al., 2024), but build no tracking-error tubes.

Funnel libraries compose controllers through containment conditions between successive funnels (Burridge et al., 1999; Tedrake et al., 2010; Majumdar and Tedrake, 2017), and temporal-logic imitation gives learned skills a related guarantee: mode-invariant, goal-reaching motion policies realize discrete plans (Wang et al., 2022). A chain of Eddies is a switched affine system (Liberzon, 2003), whose stability dwell-time conditions establish (Hespanha and Morse, 1999). These methods certify composition or stability; our chaining rule propagates a conformally calibrated

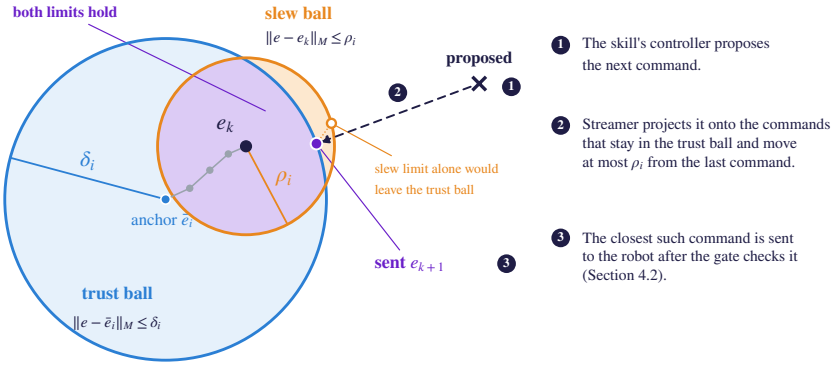


Figure 2: Each control step projects the proposed command onto the commands within the trust radius and the slew bound.

tracking-error bound through the Eddie stream, including the jump between equilibria. At the task level, language-model planners (Ren et al., 2023; Wu et al., 2025) and strong FOND synthesis for LTL_f goals (Camacho and McIlraith, 2019) choose which skill executes next; Eddies complement that layer with tracking certificates for the chosen stream.

3. Problem Formulation

A robot performs a task by chaining skills while a low-level controller tracks their commands. The hand's tracking error is the gap between where a command tells the hand (end-effector) to be and where it actually is. We want a bound on this error over the whole chain that can be checked before and during execution, so the robot stops before leaving its tolerance, and learned for a new task with no calibration set.

At step k the controller receives a command $e_k \in \mathbb{R}^m$ that sets the hand pose and the hand shape, such as the gripper aperture or finger synergies. If the controller accepts one, the command also sets a force or wrench limit. We measure only the hand pose actually achieved, $z_k \in \mathbb{R}^n$ (position and orientation), so hand shape and force limits enter only as inputs. Both live in a normalized chart of the robot's task space, the embedding space that gives Eddies their name (Appendix B).

The task allows a hand-position error of at most μ, and $S \in \mathbb{R}^{3\times n}$ extracts the position from z by picking its position rows and scaling them to metres. Fixed diagonal weights $M \succ 0$ on e and $W \succ 0$ on z, set once per robot (Appendix B), define $\|v\|_M = (v^\top M v)^{1/2}$; $\|\cdot\|$ is the Euclidean or the spectral norm.

Each robot has its own vocabulary of primitives (reach, grasp, insert, ...), shared by all its tasks (Table C.1). A skill is a primitive bound to a frame, and a task is a sequence of skills, chosen at run time by a supervisor from an LTL_f specification (Fig. 1). While skill i runs, Streamer keeps every command within a trust radius δ_i of the skill's anchor $\bar{e}_i$ (its nominal command) and limits its change per step to a slew bound ρ_i (Fig. 2):

$$\|e_k - \bar{e}_i\|_M \le \delta_i, \qquad \|e_{k+1} - e_k\|_M \le \rho_i. \tag{1}$$

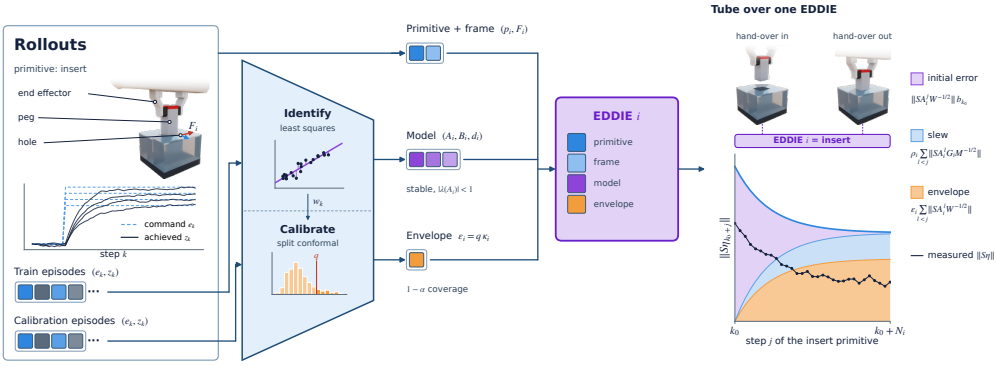


Figure 3: An Eddie pairs a model identified from rollouts with an envelope from split conformal calibration; together they give its tube.

4. Eddies

4.1. What an Eddie carries

Definition 1 *An Eddie i is a primitive bound to a frame (e.g. a grasp bound to the peg), together with a linear closed-loop model (A_i, B_i, d_i) of how the robot tracks it and an envelope $\epsilon_i \geq 0$ on the model's residual, all in one token: while i runs,*

$$z_{k+1} = A_i z_k + B_i e_k + d_i + w_k, \qquad \|W^{1/2} w_k\| \leq \epsilon_i. \tag{2}$$

The model describes the closed loop, robot plus low-level controller: $A_i \in \mathbb{R}^{n\times n}$ is the loop's memory (its eigenvalues set how fast the hand settles), $B_i \in \mathbb{R}^{n\times m}$ how the command moves the next pose, $d_i \in \mathbb{R}^n$ a constant offset such as gravity or a steady contact force, and $w_k \in \mathbb{R}^n$ the residual, what the model misses. The envelope is not a hard bound: it holds with probability at least $1-\alpha$ over episodes (3). With it, the Eddie checks each of its commands before it is sent (Section 4.2).

Fig. 3 shows how an Eddie is built: its model is fitted by least squares on the robot's training episodes that use it, regularized toward a well-excited reference Eddie of the same robot (Appendix B). If A_i is stable, a constant command e settles at $z_i^\star(e) = (I - A_i)^{-1}(B_i e + d_i)$. With $i(k)$ the Eddie running at step k, the tracking error is $\eta_k = z_k - z^\star_{i(k)}(e_k)$, the distance to where the model says the hand settles under the current command. On the Franka, this equilibrium lies a median 0.2 cm from the command itself, and the error to the command stays within the Eddie's tube on 99.9% of its test steps (Appendix D).

The envelope comes from split conformal prediction. Each Eddie carries a normalizer $\kappa_i > 0$ (the 90th percentile of its largest residual over training episodes); an episode's score is its largest normalized residual, $s = \max_k \|W^{1/2} w_k\| / \kappa_{i(k)}$, so one level q calibrated on whole episodes covers every Eddie of the stream. For a miscoverage level $\alpha \in (0,1)$ and N calibration episodes with sorted scores $s_{(1)} \leq \cdots \leq s_{(N)}$, the envelopes are

$$\epsilon_i = q\,\kappa_i, \qquad q = s_{(\lceil (N+1)(1-\alpha) \rceil)}. \tag{3}$$

If a task's episodes are exchangeable, a new episode keeps every residual within its envelope with probability at least $1-\alpha$ (Vovk, 2012; Cleaveland et al., 2024) (Fig. 4a).

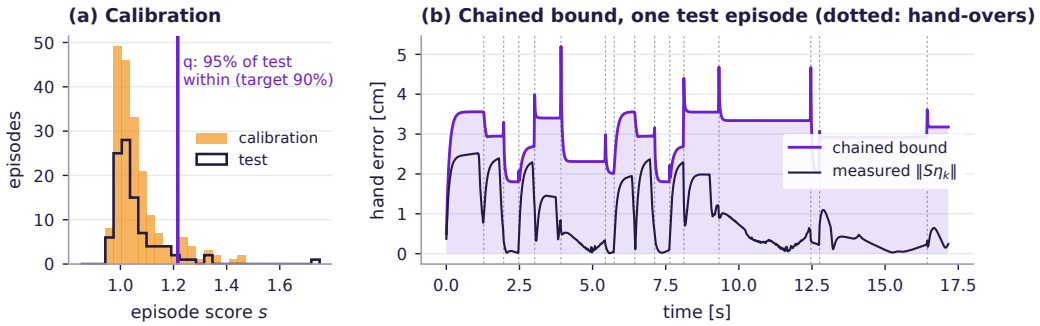


Figure 4: (a) Staged insertion: q of (3) keeps 95% of the first 100 test episodes in the envelope (94% of 200; target 90%). (b) One episode: the chained bound of Theorem 1 (violet) stays above the measured error, jumping by $J_{ii'}$ at hand-overs (dotted).

On this event, unrolling the error dynamics of the model (Hirano and Tsukamoto, 2026; Wei et al., 2025) bounds this error j steps after Eddie i starts at step k_0:

$$\|S\eta_{k_0+j}\| \le \|SA_i^j W^{-1/2}\|\,\|W^{1/2}\eta_{k_0}\| + \rho_i \sum_{l<j} \|SA_i^l G_i M^{-1/2}\| + \epsilon_i \sum_{l<j} \|SA_i^l W^{-1/2}\|, \qquad (4)$$

with $G_i = (I - A_i)^{-1} B_i$ the steady-state gain. The three terms are the decay of the initial error, the lag behind a command that moves by up to ρ_i per step, and the accumulated residual. Both claims are standard; Appendix E proves them for our score (Lemma E.1, Proposition E.2).

4.2. The Eddie's gate and chaining

At a hand-over the error does not simply carry over: it also jumps by the gap between the two Eddies' equilibria. Each Eddie therefore checks every command before it is sent, jump included, with its current model and the envelope $\epsilon_i = \max(q_t, s)\,\kappa_i$, where q_t is the coverage level of episode t, shared by the task's Eddies, $s \ge 0$ is the episode's score so far, which widens the envelope when the episode runs worse than calibrated, and κ_i is refit with the model. Before sending e_{k+1}, the running Eddie i predicts the next tracking error with its own model and the equilibrium of the Eddie i' that will run step $k+1$. Inside an Eddie $i' = i$, so the next Eddie enters only at a hand-over:

$$\begin{aligned} &\hat{\eta}_{k+1} = A_i z_k + B_i e_k + d_i - z^\star_{i'}(e_{k+1}), \\ &\text{send } e_{k+1} \text{ only if} \quad \|S\hat{\eta}_{k+1}\| + \epsilon_i \,\|SW^{-1/2}\| \le \mu. \end{aligned} \qquad (5)$$

The gate also stops the robot if the measured error $\|S\eta_k\|$ already exceeds μ; a stop sends no further command, the robot holds its last one, and the run ends unfinished unless the check passes again, as on the real robot (Section 5.5).

Every Eddie starts from the best prior there is (its own calibrated certificate, else its primitive's, else one model of the whole robot), and at each hand-over Algorithm 1 refits its model toward this prior by ridge regression on the task's data so far, kept only if stable (Appendix B). After every episode, the algorithm moves the coverage level by the quantile form of adaptive conformal inference (Gibbs and Candès, 2021; Angelopoulos et al., 2023):

$$q_{t+1} = q_t + \gamma\,(\mathrm{err}_t - \alpha), \qquad \mathrm{err}_t = \mathbf{1}\{s_t > q_t\}, \qquad (6)$$

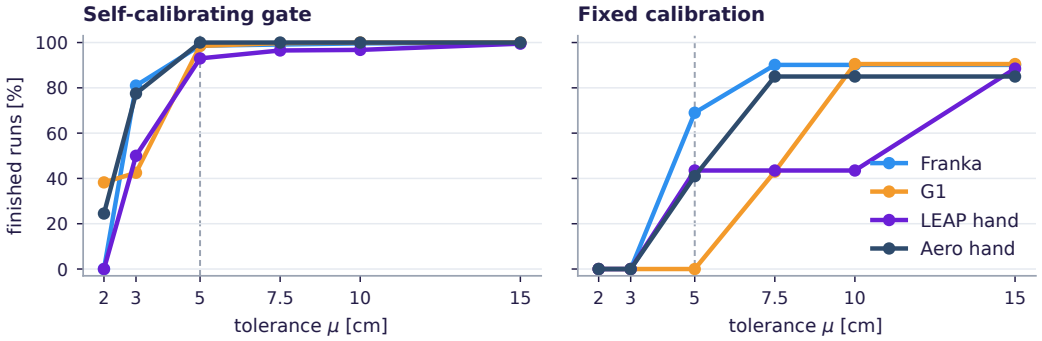


Figure 5: Finished runs against tolerance for both gates from the task's own calibration; one curve per robot (200 test episodes per task).

where $\gamma > 0$ is a step size, 0.5 in our experiments, and s_t is the score of episode t under the current models and normalizers.

Algorithm 1: Self-calibrating gate, episode t

Input: models and normalizers of the Eddies (priors), coverage level q_t, tolerance μ
$s \leftarrow 0$;
for *each step k, with running Eddie i* **do**
  if $\|S\eta_k\| > \mu$ ***or*** (5) *fails* **then** stop the robot;
  send e_{k+1} and measure z_{k+1};
  $s \leftarrow \max\left(s,\ \|W^{1/2} w_k\| / \kappa_i\right)$; `// episode score so far`
  $\epsilon_i \leftarrow \max(q_t, s)\,\kappa_i$; `// widen the envelope if needed`
  if *a hand-over follows* **then** refit $(A_i, B_i, d_i, \kappa_i)$ by ridge regression toward its prior;
end
$q_{t+1} \leftarrow q_t + \gamma\,(\mathbf{1}\{s > q_t\} - \alpha)$; `// adaptive conformal update`

Our guarantees rest on two assumptions. **(A1)** *Exchangeable episodes:* the episodes of a task come from independent scene draws, while steps within an episode may be correlated. **(A2)** *Stable models:* every model the gate uses has $\max_\lambda |\lambda(A_i)| < 1$. We certify only Eddies that satisfy it.

Theorem 1 (Certified Eddie streams) *Let (A2) hold and let the commands satisfy* (1)*. In the embedding space with metric W, let $b_k = \|W^{1/2}\eta_k\|$ be the tracking error and $a_i = \|W^{1/2} A_i W^{-1/2}\|$ the one-step gain of Eddie i. Let $J_{ii'} = \|W^{1/2}\big(z_i^\star(e_k) - z_{i'}^\star(e_{k+1})\big)\|$ be the jump between the equilibria when Eddie i hands over to Eddie i' after step k.*

(a) Chained certificate. *Under (A1) and with the envelopes* (3)*, with probability at least $1-\alpha$ over episodes, at every step: within each Eddie the error follows the tube* (4) *from its value when the Eddie starts, and across each hand-over*

$$b_{k+1} \le a_i\, b_k + \epsilon_i + J_{ii'}. \tag{7}$$

Chained from the measured error at the first step, these bounds hold for $\|S\eta_k\|$ over the whole Eddie stream and are finite for every finite stream.

(b) Gate. *For any stable models, without (A1): whenever the residual of the running Eddie stays within its envelope, the command its gate* (5) *sends keeps $\|S\eta_{k+1}\| \le \mu$, and every run that finishes stays within μ.*

(c) Miss rate. *Under* (6)*, with scores and q_1 in $[0, s_{\max}]$, $|\frac{1}{T}\sum_{t\le T} \mathrm{err}_t - \alpha| \le (s_{\max} + \gamma)/(\gamma T)$ for any sequence of episodes and tasks.*

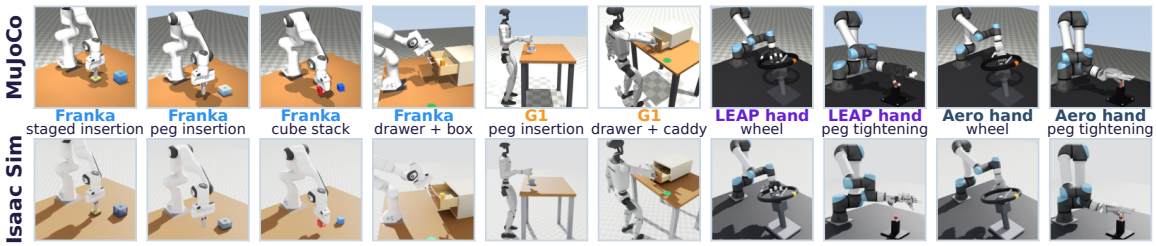


Figure 6: The ten tasks in MuJoCo (top) and Isaac Sim (bottom) from the same task code; the LEAP and Aero hands sit on a UR5e arm.

Part (a) needs no contraction across hand-overs (81 of the 85 certified Eddies have $a_i < 1$; Appendix B) and assumes the fixed models of (3); (b) and (c) hold for any stable models, also after Algorithm 1 refits them (Proposition E.3).

4.3. Vocabulary and specifications

The Eddies of a task share one calibrated coverage level; a primitive added later gets a fixed share of the error budget, so adding it never changes an existing certificate (Proposition E.4; Section 5.4). Each primitive also carries a symbolic contract (precondition and possible outcomes). Given an LTL_f goal (De Giacomo and Vardi, 2013), Streamer solves a product game over the Eddies; its strategy is the Eddie stream (order, recoveries and when to stop), the goal is refused before motion if none exists, and every Eddie on every branch goes through the gate (Appendix A: the projection onto (1) is well-posed and the supervisor finishes or refuses).

5. Experiments

We test Eddies on ten MuJoCo tasks with four robots (Todorov et al., 2012; Zakka et al., 2022). Each task uses only its robot's primitives (Fig. 6). The Franka does peg insertion, cube stacking, drawer opening with a box transfer, and staged insertion (16 Eddies: pick the peg, stage it on a fixture, regrasp, insert at 1–2 mm clearance, push home). The G1 humanoid does peg insertion and drawer opening with a caddy transfer, and a UR5e with a LEAP hand and with an Aero hand each turns a wheel and tightens a peg. Staged insertion, peg insertion and cube stacking are also certified in Isaac Sim from the same code (NVIDIA, 2026).

Each task has 200 training, 200 calibration and 200 test episodes, $\alpha = 0.1$, and a certificate is violated if the measured hand position leaves its tube. Gates are replayed on the logged test episodes, so the outcome of every stopped run without the gate is known.

5.1. Certificates on four robots

The certificates hold: with the calibration fixed, tubes are violated in 5 of 2,000 test episodes, and most are 3–5 cm wide (Table D.1). Envelopes are missed more often, in 6–16.5% of Franka test episodes (Table D.3; the drawer's 16.5% is within sampling noise, Appendix D), and in 8.5–10.5% under the self-calibrating gate, close to α as (6) intends. At 5 cm, the recursion of Theorem 1 certifies 77–100% of the hand-overs of every Franka task, against 69–86% for funnel containment; whole episodes certify a priori on peg insertion at 5 cm and on three tasks from 7.5 cm (Table D.4).

Checking one command at a time more than doubles the finished runs (Fig. 5): at 5 cm with the calibration fixed, checking every planned Eddie's tube before motion (each hand-over from the

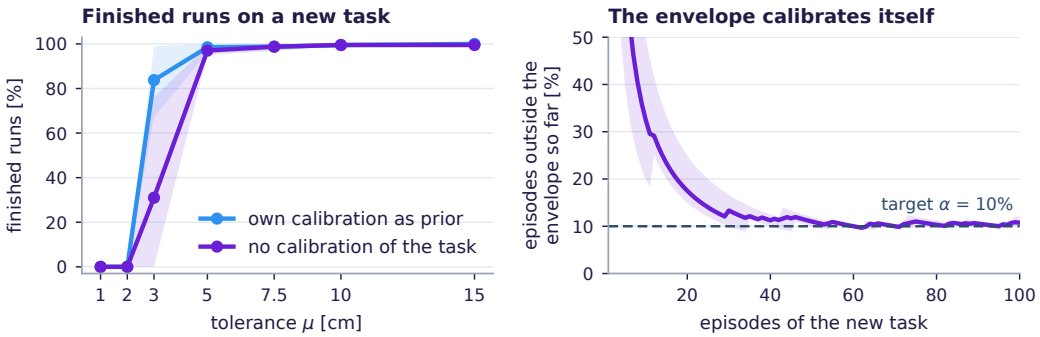


Figure 7: Four Franka tasks held out of training, no calibration: finished runs (left), share of episodes so far outside the envelope (right); mean and range.

Table 1: Our certificate on different tokens of each robot's data: BeT, FAST, VQ-BeT and a global model as in Nath et al. (2026), BeT and FAST with as many tokens as the robot has Eddies (35 Franka, 13 per hand, 25 G1); the last two rows ablate Eddies' models and calibration. Tube p90 [cm] and runs finished of 400 test runs, calibration fixed, at $\mu = 5$ / 7.5 cm (G1: 7.5 / 10 cm, as its balance control widens the tubes, p90 up to 7.9 cm, Table D.1, and no plan fits within 5 cm with any tokens). Bold: not significantly below the best (paired bootstrap, 95% intervals of about ± 10 runs; Appendix D).

	Franka		LEAP hand		Aero hand		G1	
Tokens	tube	finished 5 / 7.5 cm	tube	finished 5 / 7.5 cm	tube	finished 5 / 7.5 cm	tube	finished 7.5 / 10 cm
Eddies	**4.3**	**275 / 359**	**11.9**	**174 / 174**	**6.5**	**164 / 340**	**7.9**	**172 / 362**
One global model	8.5	0 / 262	22.0	0 / 0	24.4	0 / 0	79.3	0 / 0
One model per primitive	8.1	0 / 264	12.5	0 / 0	12.0	0 / 170	15.7	0 / 0
BeT	8.7	169 / 261	26.4	0 / 0	12.6	0 / 0	9.3	0 / 187
FAST + k-means	13.7	177 / 272	55.3	0 / 0	16.2	0 / 0	14.9	0 / 0
VQ-BeT	16.0	264 / 264	18.2	0 / 0	21.0	0 / 0	10.2	0 / 0
Eddies, one reference model	18.4	0 / 0	17.6	0 / 0	10.5	0 / 0	60.1	**172** / 172
Eddies, per-step bound	–	0 / 0	–	0 / 0	8.7	0 / 0	–	0 / 0

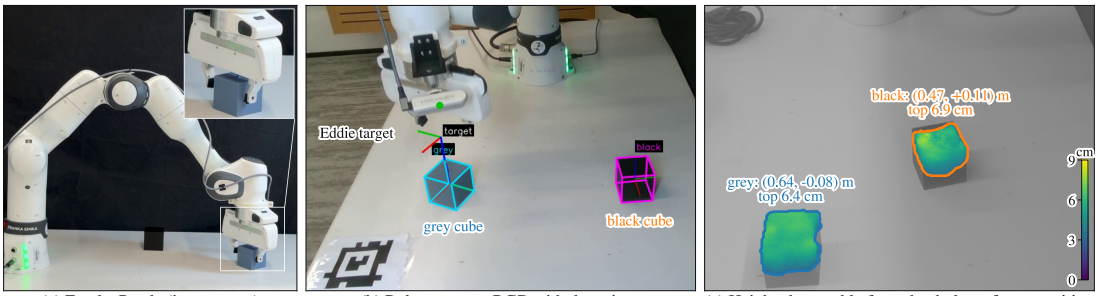
measured error) finishes 890 of 2,000 runs, checking only the next command 1,968, and the self-calibrating gate, which adds the refits and the updates of q, 1,955 (98%) from the same calibration (two different guarantees: the plan check certifies the stream before motion, the one-step check each command from the measured state); its benefit shows on new tasks, which start with no calibration (Section 5.4). At 2 cm, 1,627 runs would leave μ, and the look-ahead stops 97% of them before they do, where a threshold on the measured error alone stops none in time. With every Eddie 1.5–4 times faster, the tasks still succeed, but 35–100% of runs would leave μ at 5 cm; with the calibration fixed, the gate refuses every plan at these speeds before motion (Table D.5).

5.2. Eddies against other tokens

Published tokenizers carry no model or envelope, so each gets our certificate on the same data of each robot (Table 1), and any difference in tightness comes from the token itself. Eddies are tightest on every robot and finish the most runs; on the hands and the G1 no other tokens finish a run at the smaller tolerance. Token counts are matched: BeT with twice the tokens comes level on the Franka, at twice the hand-overs per episode (Appendix D). Only Eddies, labelled by primitive and frame, have symbolic contracts and a plan known before motion.

Table 2: Certificates borrowed from a source system or re-identified on the target (Isaac Sim: refit on the MuJoCo data plus 10 Isaac Sim episodes per task, calibrated on 50 more); viol.: test episodes that leave a tube; finished: runs the gate finishes at $\mu = 15$ cm; bold: the better per task, both when equal.

			borrowed		re-identified	
Source → target	Task	episodes	viol.	finished	viol.	finished
LEAP → Aero	Wheel	200	20	78	**0**	**176**
	Peg tightening	200	5	0	**0**	**164**
Aero → LEAP	Wheel	200	**0**	22	1	**180**
	Peg tightening	200	**0**	0	**0**	**174**
MuJoCo → Isaac Sim	Staged insertion	50	10	22	**0**	**45**
	Peg insertion	50	5	19	**0**	**46**



(a) Franka Panda (inset: grasp) (b) Robot camera, RGB with detections (c) Height above table from depth; base-frame position

(d) reach: cube pushed (e) reach: re-targets (f) certified stop (g) reach: resumes (h) grasp (i) place: stacked

Figure 8: Cube stacking on the real Franka. (a) The Panda grasping the grey cube (inset). (b) Robot-camera RGB feed with the detected cubes and the active Eddie's target. (c) Cube heights from the same camera's depth, base-frame positions. (d–i) Under perturbation: a stick pushes the cube and the arm re-targets; out of the safe set, the arm holds in a certified stop until the cube is back, then grasps and stacks it.

Both the calibration and each Eddie's own model matter: a per-step envelope with a union bound (Hirano and Tsukamoto, 2026) certifies no plan at these tolerances (too few calibration steps per Eddie), and the robot's reference model for every Eddie widens the tube 1.5–7.6 times (Appendix D).

5.3. Certificates across robots and engines

Table 2 moves certificates between systems that run the same Eddies (the LEAP and Aero hands on one UR5e arm; MuJoCo and Isaac Sim, Fig. C.1) and compares each borrowed certificate with one re-identified on the target. Borrowed certificates are violated or too wide to finish most runs; re-identified ones are violated at most once and finish over 80% of runs, in Isaac Sim after only ten training episodes per task.

Across robots the same primitive's certificate differs: pulling a drawer open has a 3.0 cm tube on the Franka and 8.3 cm on the G1, while reaching for a peg stays at 2.9–4.9 cm on all four (Table D.6).

Table 3: Cube stacking in MuJoCo (first 100 test episodes), in Isaac Sim and on the real Franka at $\mu = 5$ cm. Both gates see the same runs from the certificate shown (same seeds): replayed on the logs in simulation, online on the robot, where half of the 50 runs are perturbed (fast runs replayed). Viol.: runs that leave a tube; tube p90: 90th percentile of the tube radius over executed steps; fin./stop.: runs without / with a stop (a stopped robot run holds until the check passes again); caught: runs at 2–4$\times$ speed refused or stopped before leaving μ, of those that would.

						fixed gate		self-calibrating gate	
System	Certificate	runs	success	viol.	tube p90 [cm]	fin. / stop.	caught	fin. / stop.	caught
MuJoCo	MuJoCo	100	99	0	4.3	89 / 11	298 / 298	96 / 4	287 / 298
Isaac Sim	MuJoCo	50	50	0	4.3	31 / 19	150 / 150	48 / 2	148 / 150
	re-identified	50	50	0	4.0	41 / 9	150 / 150	48 / 2	147 / 150
robot	MuJoCo	50	50	0	4.3	30 / 20	150 / 150	46 / 4	145 / 150
	re-identified	50	50	0	4.1	43 / 7	150 / 150	45 / 5	143 / 150

5.4. New tasks, primitives and specifications

A new task reuses the Eddies it shares with other tasks, falls back to primitive or global models for the rest and tunes itself as it runs (Fig. 7; Table D.2). With no task episodes, the self-calibrating gate finishes nearly every run at 5 cm with 1–5 false stops; the conformal certificate finishes none without calibration and needs about 55 episodes to finish 83–93.

A new push primitive certifies every staged-insertion Eddie and leaves the nine existing primitives' certificates unchanged. From LTL_f goals, the product game recovers all four hand-written Franka plans, runs five new goals (99–100% success) and refuses three impossible ones before motion; a network trained on the supervisor's choices (92–100% of held-out hand-overs right) emits the Eddies itself with the supervisor's 99–100% nominal success (Appendix D).

5.5. Real experiments

Cube stacking (eight Eddies) also runs in Isaac Sim and on a real Franka Panda (Fig. 8), where the 1 kHz compliant controller of CRISP (San José Pro et al., 2026) tracks the 50 Hz Eddie commands. Both gates run online on the same test runs, from the MuJoCo certificate or from one refit on the MuJoCo data plus 50 of the robot's own episodes, calibrated on 50 more (Table 3).

Unlike for insertion, the MuJoCo certificate holds on both systems: no run leaves a tube, and every Isaac Sim and robot run stacks the cube; its envelopes are missed more often, so the fixed gate stops 19 of 50 runs in Isaac Sim and 20 on the robot (9 and 7 once re-identified), the self-calibrating gate 2–5. At two to four times the speed (run ungated, the gates replayed as in simulation), the fixed gate catches every run that would leave μ and the self-calibrating gate 95–99%.

A cube pushed during the approach is followed by the arm; pushed out of the safe set (the cube poses for which the reach Eddie's check (5) passes, from its certificate), it makes the Eddie refuse its next command, a certified stop that holds the arm for 15 s until the cube is back, and the run still ends with the cube stacked (Fig. 8d–i).

6. Conclusion

Eddies are action tokens that carry the dynamics of the robot that runs them; they chain into a certificate for the whole task, gate their own commands and calibrate themselves as they run, in simulation and on a real Franka. The certificate covers tracking, not contact outcomes; the models

are linear per Eddie and the contracts are written by hand. A learned proposer already emits Eddies (Section 5.4), as a vision-language-action model could.

Generative AI Use

Generative AI tools were used for writing and editing. The authors reviewed and verified all AI-assisted content and take responsibility for the final content of the paper.

References

Christopher Agia, Rohan Sinha, Jingyun Yang, Ziang Cao, Rika Antonova, Marco Pavone, and Jeannette Bohg. Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress. In *Conference on Robot Learning (CoRL)*, volume 270 of *Proceedings of Machine Learning Research*, pages 689–723, 2024.

Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, et al. Do as I can, not as I say: Grounding language in robotic affordances. In *Conference on Robot Learning (CoRL)*, volume 205 of *Proceedings of Machine Learning Research*, pages 287–318, 2022.

Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. Conformal PID control for time series prediction. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning (CoRL)*, volume 229 of *Proceedings of Machine Learning Research*, pages 2165–2183, 2023.

R. R. Burridge, A. A. Rizzi, and D. E. Koditschek. Sequential composition of dynamically dexterous robot behaviors. *The International Journal of Robotics Research*, 18(6):534–555, 1999.

Alberto Camacho and Sheila A. McIlraith. Strong fully observable non-deterministic planning with LTL and LTL$_f$ goals. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, pages 5523–5531, 2019.

Matthew Cleaveland, Insup Lee, George J. Pappas, and Lars Lindemann. Conformal prediction regions for time series using linear complementarity programming. In *AAAI Conference on Artificial Intelligence*, volume 38, pages 20984–20992, 2024. doi: 10.1609/aaai.v38i19.30089.

Murtaza Dalal, Deepak Pathak, and Ruslan Salakhutdinov. Accelerating robotic reinforcement learning via parameterized action primitives. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.

Giuseppe De Giacomo and Moshe Y. Vardi. Linear temporal logic and linear dynamic logic on finite traces. In *International Joint Conference on Artificial Intelligence (IJCAI)*, pages 854–859, 2013.

Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 1660–1672, 2021.

João P. Hespanha and A. Stephen Morse. Stability of switched systems with average dwell-time. In *IEEE Conference on Decision and Control (CDC)*, pages 2655–2660, 1999.

Koki Hirano and Hiroyasu Tsukamoto. Conformal Koopman for embedded nonlinear control with statistical robustness: Theory and real-world validation. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. arXiv:2603.21580.

Seungjae Lee, Yibin Wang, Haritheja Etukuru, H. Jin Kim, Nur Muhammad Mahi Shafiullah, and Lerrel Pinto. Behavior generation with latent actions. In *Proceedings of the International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 26991–27008, 2024.

Daniel Liberzon. *Switching in Systems and Control*. Birkhäuser, Boston, 2003.

Lars Lindemann, Matthew Cleaveland, Gihyun Shim, and George J. Pappas. Safe planning in dynamic environments using conformal prediction. *IEEE Robotics and Automation Letters*, 8(8): 5116–5123, 2023. doi: 10.1109/LRA.2023.3292071.

Lars Lindemann, Yiqi Zhao, Xinyi Yu, George J. Pappas, and Jyotirmoy V. Deshmukh. Formal verification and control with conformal prediction: Practical safety guarantees for autonomous systems. *IEEE Control Systems Magazine*, 45(6):72–122, 2025. doi: 10.1109/MCS.2025.3611545.

Anirudha Majumdar and Russ Tedrake. Funnel libraries for real-time robust feedback motion planning. *The International Journal of Robotics Research*, 36(8):947–982, 2017. arXiv:1601.04037.

Atharva Mete, Haotian Xue, Albert Wilcox, Yongxin Chen, and Animesh Garg. QueST: Self-supervised skill abstractions for learning continuous control. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.

Soroush Nasiriany, Huihan Liu, and Yuke Zhu. Augmenting reinforcement learning with behavior primitives for diverse manipulation tasks. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 7477–7484, 2022. doi: 10.1109/ICRA46639.2022.9812140.

Devesh Nath, Haoran Yin, and Glen Chou. Scalable data-driven reachability analysis and control via Koopman operators with conformal coverage guarantees. In *Learning for Dynamics and Control Conference (L4DC)*, volume 331 of *Proceedings of Machine Learning Research*, pages 71–97, 2026.

NVIDIA. Isaac Sim 6.0 release notes. https://docs.isaacsim.omniverse.nvidia.com/6.0.0/overview/release_notes.html, 2026. Accessed Sep. 19, 2026.

Karl Pertsch, Kyle Stachowicz, Brian Ichter, Danny Driess, Suraj Nair, Quan Vuong, Oier Mees, Chelsea Finn, and Sergey Levine. FAST: Efficient action tokenization for vision-language-action models. In *Robotics: Science and Systems (RSS)*, 2025. doi: 10.15607/RSS.2025.XXI.012.

Allen Z. Ren, Anushri Dixit, Alexandra Bodrova, Sumeet Singh, Stephen Tu, Noah Brown, et al. Robots that ask for help: Uncertainty alignment for large language model planners. In *Conference on Robot Learning (CoRL)*, volume 229 of *Proceedings of Machine Learning Research*, pages 661–682, 2023.

Daniel San José Pro, Oliver Hausdörfer, Ralf Römer, Maximilian Dösch, Martin Schuck, and Angela P. Schoellig. CRISP – compliant ROS2 controllers for learning-based manipulation policies and teleoperation. *IEEE Robotics and Automation Practice*, 1:103–107, 2026. doi: 10.1109/RAP.2026.3690413.

Nur Muhammad Mahi Shafiullah, Zichen Jeff Cui, Ariuntuya Altanzaya, and Lerrel Pinto. Behavior transformers: Cloning k modes with one stone. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Russ Tedrake, Ian R. Manchester, Mark Tobenkin, and John W. Roberts. LQR-trees: Feedback motion planning via sums-of-squares verification. *The International Journal of Robotics Research*, 29(8):1038–1052, 2010.

E. Todorov, T. Erez, and Y. Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5033, 2012. doi: 10.1109/IROS.2012.6386109.

Vladimir Vovk. Conditional validity of inductive conformal predictors. In *Asian Conference on Machine Learning (ACML)*, volume 25 of *Proceedings of Machine Learning Research*, pages 475–490, 2012.

Yanwei Wang, Nadia Figueroa, Shen Li, Ankit Shah, and Julie Shah. Temporal logic imitation: Learning plan-satisficing motion policies from demonstrations. In *Proceedings of the Conference on Robot Learning (CoRL)*, volume 205 of *Proceedings of Machine Learning Research*, pages 94–105, 2022.

Sihang Wei, Melkior Ornik, and Hiroyasu Tsukamoto. Conformal contraction for robust nonlinear control with distribution-free uncertainty quantification. In *IEEE Conference on Decision and Control (CDC)*, pages 3502–3507, 2025. doi: 10.1109/CDC57313.2025.11312813.

Yi Wu, Zikang Xiong, Yiran Hu, Shreyash S. Iyengar, Nan Jiang, Aniket Bera, Lin Tan, and Suresh Jagannathan. SELP: Generating safe and efficient task plans for robot agents with large language models. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

K. Zakka, Y. Tassa, and MuJoCo Menagerie contributors. MuJoCo Menagerie: A collection of high-quality simulation models for MuJoCo. https://github.com/google-deepmind/mujoco_menagerie, 2022.

Hongyi Zhou, Weiran Liao, Xi Huang, Yucheng Tang, Fabian Otto, Xiaogang Jia, et al. BEAST: Efficient tokenization of B-splines encoded action sequences for imitation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2506.06072.